

Marco Castellari

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16/09/1993 – Imola (BO)

PROFILE

Versatile Electronics and Telecommunications Engineer with expertise in firmware and software design and development for high-performance systems in motorsport, industrial automation and medical devices. Adaptable team player with proven ability to deliver under pressure, with a passion for new challenges and continuous learning.

WORK EXPERIENCE

R&D Software Engineer Oct 2024 – Present
Cefla S.C., Imola (IT)

- Software architecture and real-time C++ framework development for X-ray imaging systems
- Design and implementation of medical imaging reconstruction algorithms

R&D HW/FW Engineer May 2023 – Sept 2024
Mectronik s.r.l., Imola (IT)

- Firmware development for high-performance motorsport applications and collaboration in hardware design
- Trackside engineering support for race strategy and engine map optimization

R&D Software Engineer Apr 2018 – May 2023
SACMI Innovation Lab, Imola (IT)

- Development of C++/GPU and FPGA solutions for high-performance and real-time computing
- Maintenance of legacy C applications and on-site prototype commissioning

EDUCATION

State examination with qualification for the profession of Information Engineer 2021
Alma Mater Studiorum - Università di Bologna

MSc Electronics & Telecommunications Eng. 2015–2018

Università degli Studi di Ferrara 110/110 *cum Laude*
Thesis: *Performance Evaluation of Emerging FPGA Technologies*

BSc Electronics & Computer Science Eng. 2012–2015
Università degli Studi di Ferrara 110/110
Thesis: *Localization Systems based on LTE Signals of Opportunity*

High School Diploma 2007 - 2012
ITIS - G. Marconi, Lugo (RA) 100/100 *cum Laude*
QUALIFIED INDUSTRIAL TECHNICIAN
Specialization in Electronics and Telecommunications

RESEARCH EXPERIENCE

Fermilab Summer Internship July – September 2017
Fermi National Accelerator Laboratory, Batavia (USA, IL)
Novel RF Tuning of Jacketed Multi-Cell SRF Cavities

MPSoC Research Group August – September 2015
Università degli Studi di Ferrara
Optical Interconnection Networks for HPC

SKILLS

Programming C/C++, Qt, CUDA, Python, MATLAB, VHDL, Linux, Windows

Hardware STM32, ESP32, NXP, FPGA, CAN bus, Lauterbach, Segger, Laboratory Instrumentation

Tools Git/SVN, Docker, Jenkins, Altium Designer, Vivado, Quartus

Architecture High-Performance Computing, Real-Time and Embedded Systems

Soft Skills Passion for learning, Care for details, Teamwork

Languages Italian (Native), English

INTERESTS

Electronics for motorsport (suspension testbench and telemetry system) • Mechanics and engines • Firmware/Software conference attendance • Travelling

PUBLICATION

Tuning of Multicell Superconducting Accelerating Cavities using Pressurized Balloons, 2022 IEEE Transactions on Applied Superconductivity.

Correlating Power Efficiency and Lifetime to Programming Strategies in RRAM-Based FPGAs, 2018 New Generation of CAS (NGCAS).

Populating and exploring the design space of Wavelength-Routed Optical Network-on-Chip topologies by leveraging the add-drop filtering primitive, 2016 Tenth IEEE/ACM International Symposium on Networks-on-Chip (NOCS).